

ZQE: Applications and Practical Recipe

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Abstract

This note summarizes practical applications and an actionable recipe for ZQE (“Zeroth-order Quadratic Estimator”) as implemented in the project. The document is a concise, self-contained briefing intended for a research supervisor or collaborator: motivating idea, estimator definition, recommended practical settings, and where ZQE is especially useful.

1. One-Line Summary

ZQE (“Consistent Decoder Estimation Without Accurate Posterior Inference”) is a simple, simulation-based objective that estimates decoder parameters (loading matrix W and intercept b) directly by matching decoder outputs under real and simulated data using a fixed imputation of latent variables. It enables consistent recovery of interpretable decoder structure even when accurate posterior inference is difficult.

2. Formal Definition (short)

Let $h_{\text{enc}}(y)$ and $h_{\text{dec}}(y)$ be chosen sufficient-statistic transforms of the observed response y . Let $\eta(z) = Wz + b$ be the linear predictor. The ZQE cross-term objective optimises

$$\mathcal{L}_{\text{ZQE}}(W, b) = -\left(\frac{1}{N} \sum_{i=1}^N h_{\text{dec}}(y_i)^\top \eta(\hat{z}(y_i)) - \frac{1}{M} \sum_{m=1}^M h_{\text{dec}}(y_m^{\text{sim}})^\top \eta(\hat{z}(y_m^{\text{sim}}))\right),$$

where the imputed latent is the Gaussian MAP closed-form:

$$\hat{z}(y) = (W^\top W + \lambda I)^{-1} W^\top (h_{\text{enc}}(y) - b).$$

The outer optimisation updates (W, b) (e.g. by SGD) while treating $\hat{z}(\cdot)$ as a fixed imputation (no gradient through the encoder). A centering term is estimated by simulation from the generative model to remove nuisance terms.

3. Why this helps (intuition)

- By fixing the imputation, gradients target only decoder parameters, avoiding the complexity of simultaneous posterior learning. - The centering (simulation) term cancels biases due to the chosen h -transforms and makes the objective robust across link choices. - Closed-form Gaussian MAP encoder is cheap and stable for linear decoders, and allows interpretable sparsity recovery in W .

4. Practical Recipe (recommended defaults)

- Transform choices: use $h(\cdot) = \log(1 + y)$ (“log1p”) or the Anscombe/sqrt transforms for low counts; choose encoder/decoder transforms independently to match modelling assumptions.
- Warm start: initialise W from the top- Q left singular vectors of centered $h(Y)$ (SVD warm start). This places parameters in a principal component rotation, accelerating sparsification.
- Proximal sparsity: apply an *LR-aware* proximal L1 operator (soft-threshold after each optimiser step) with threshold $\tau = \lambda \cdot \text{lr}$. This produces exact zeros, and scaling with the current learning rate makes sparsification aggressive early and fine later.
- Centering draws: draw a large number of simulated samples per outer update (we recommend at least $M \gtrsim 10N$ or a fixed minimum such as $M = 2000$) under `torch.no_grad()`*[this cheaply reduces centering variance without affecting parameter gradients. Numerical stability : add a small jitter to the encoder linear system, e.g. $W^\top W + (\lambda + \varepsilon)I$ with $\varepsilon \approx 10^{-6}$, and detect any fully-zero loading columns after proximal steps and reinitialise them with tiny random noise to avoid singular solves.*

5. Typical hyperparameters to try

Start with these sensible defaults and then sweep if necessary:

- Optimiser: SGD with an initial learning rate of ≈ 1.0 and a Robbins–Monro decay schedule (flat-phase then power decay).
- Proximal L1: $\lambda \in [0.001, 0.05]$ (larger values make sparser W).
- Centring draws: $M = \max(10N, 2000)$.
- Ridge (encoder): $\lambda_{\text{ridge}} \approx 1.0$ plus jitter 10^{-6} .

6. Applications

ZQE is particularly useful where the decoder/observation model is the primary object of inference and where posterior accuracy is costly or unnecessary:

- Sparse factor loadings recovery (e.g. interpretable latent factors in single-cell RNA-seq or ecological trait models).
- High-dimensional GLLVMs where posterior inference is expensive and a simple MAP imputation suffices to estimate a consistent decoder.
- Topic or latent factor models where direct decoder estimation (topics, item factors) is preferred to full posterior amortisation.
- Recommender or choice models where interpretable loadings are the primary target and exact posterior uncertainties are secondary.

7. Caveats and recommendations

- ZQE recovers the decoder under modelling assumptions on h and the encoder; poor choices of transforms can slow or bias recovery — use simple transforms ($\log 1p$, $\sqrt{\cdot}$) as baselines.
- If columns collapse to exact zero early (over-aggressive prox), add small reinitialisation and reduce proximal strength or learning rate.
- Use SVD warm starts when seeking sparse solutions — random initialisation may require longer runs and more tuning.

8. How to compile

On a machine with a standard LaTeX distribution (`pdflatex`):

```
pdflatex paper/pitch/zqe_applications.tex
```

If you are on Windows and do not have LaTeX, installing MiKTeX is recommended. On Linux, install TeX Live (e.g. `sudo apt install texlive-latex-recommended`).

9. Next steps / Offer

If you would like, I can:

- run a quick compile in this environment and return the PDF, or
- adapt the note to include a one-page figure from your experiment (e.g. coefficient paths from a `lambda` sweep).

[11pt]article[utf8]inputenc amsmath,amssymb geometry hyperref parskip margin=1in itleZQE: Applications and Practical Recipe (revised) Will White May 27, 2026

Abstract

This one-page brief summarises practical recommendations for the Z_q (ZQE) estimator based on the paper draft ('paper/main.tex') and the project working notes ('CLAUDE.md'). It emphasises the *Gaussian proxy* as the canonical practical choice, documents implementation safeguards observed in experiments, and provides a concise recipe for reproducible experiments and sparse recovery.

1. Core message

The Z_q estimator yields consistent decoder estimation even with misspecified or frozen surrogate inference, provided the centered estimating equations are used and a local Jacobian nonsingularity condition holds (see 'paper/main.tex'). For practical use, the Gaussian proxy (choose $T(y)$ to make a homoskedastic proxy) is the canonical instance: it is computationally trivial, theoretically tractable, and empirically robust in large- p settings.

2. Precise estimator (practical form)

Pick a transform $T(y)$ (e.g. $\log(1+y)$ or $\sqrt{y}$) and form the Gaussian proxy so that

$$T(y) \mid z \approx \mathcal{N}(Wz + b, \sigma^2 I).$$

The closed-form MAP encoder is

$$\hat{z}(y) = (W^\top W + \sigma^2 I)^{-1} W^\top (T(y) - b),$$

and the ZQE objective is the centered cross-term

$$\mathcal{L}(\theta) = -\left(\mathbb{E}_{\text{data}}[T(Y) \cdot \eta(\hat{z}(Y))] - \mathbb{E}_{\theta}[T(Y^{\text{sim}}) \cdot \eta(\hat{z}(Y^{\text{sim}}))]\right).$$

Compute the centering expectation by simulation (large fixed bank or on-the-fly MC under `no_grad()`). Do not backpropagate through the encoder solves.

3. Why the Gaussian proxy (summary of repo notes)

- It gives a closed-form encoder and often yields deterministic or low-variance centering terms (multiple imputation with fixed seeds).
- In large- p regimes the ridge encoder concentrates (LLN in p), meaning the proxy approaches the posterior mean and ZQE recovers the complete-data score.
- The Gaussian proxy lets us verify the Jacobian nonsingularity condition analytically in canonical settings (Poisson GLLVM), which strengthens the theoretical contribution for a statistical venue (JRSS-B recommendation).

4. Implementation checklist (must-do)

- **Encoder under `no_grad`:** compute $\hat{z}(\cdot)$ inside `extttorch.no_grad()` so gradients flow only through $\eta(\hat{z})$.
- **SVD warm start:** initialise W from top- Q PCs of centered $T(Y)$ to place parameters in a rotation where sparsity is discoverable.
- **LR-aware proximal L1:** apply soft-threshold after each optimiser step with threshold $\tau = \lambda \times \text{lr}_{\text{now}}$ (proximal ISTA scaled by current LR). This yields exact zeros and a meaningful regularisation path when warm-starting large $\rightarrow$ small λ .
- **Large centering budget:** draw many simulated examples for the centering term (recommend $M = \max(10N, 2000)$) inside `no_grad()` to sharply reduce Monte-Carlo noise at low cost.
- **Numerical safeguards:** add a tiny jitter $\varepsilon \approx 10^{-6}$ when forming $W^\top W + (\sigma^2 + \varepsilon)I$, and detect any fully-zero columns after prox steps and reinitialise them with tiny Gaussian noise.

5. Practical defaults and tuning

- Optimiser: SGD with flat phase + Robbins–Monro decay; initial lr 1.0.
- Proximal L1: try $\lambda \in [0.001, 0.05]$; warm-start grid from large $\rightarrow$ small.
- Encoder ridge / proxy variance: $\sigma^2 \approx 1.0$ (tune as needed).
- Centering draws: $M = \max(10N, 2000)$ as a robust default.

6. Experiments and story advice (paper/main.tex + CLAUDE.md)

- Lead with the general ZQE framework and immediately present the Gaussian proxy as the canonical, constructive instance (answers the practical "what should I use?" question). - Emphasise the large- p concentration phenomenon and report a p-sweep figure. - De-emphasise the frozen-amortised-encoder experiment as a main claim (it creates interpretational complications); keep a short caveat in discussion. - Include verification of the Jacobian nonsingularity for Poisson GLLVMs with the Gaussian proxy (this closes the main theoretical gap noted in project notes and helps target JRSS-B).

7. Common pitfalls

- Do not backpropagate through the encoder; doing so reintroduces the chain-rule terms the theory cancels and creates gradient leakage.
- L1 inside the loss (non-proximal) rarely produces exact zeros — use a proximal soft-threshold after the optimizer step to recover exact sparsity.
- If many columns die early, lower the prox strength or reinitialise dead columns — otherwise the encoder linear system becomes ill-conditioned.

8. How to compile and share

To compile locally with `pdflatex`:

```
pdflatex paper/pitch/zqe_applications.tex
```

If compilation fails, install a TeX distribution (MiKTeX on Windows, TeX Live on Linux). If you want, I can run a quick compile here and return the PDF.

9. Offer

I can also:

- insert a one-page figure (lambda-path coefficient heatmap or active-count curve) into this note, or
- produce a short appendix summarising the Jacobian verification for the Poisson GLLVM with the Gaussian proxy (ready for JRSS-B reviewers).

Best,
Will White